

Jun Yu (Johnny) Chen

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EDUCATION

UNIVERSITY OF PENNSYLVANIA

Major: MSE in Data Science, MSE in Computer & Information Science

Philadelphia

Expected May 2027

UNIVERSITY OF CALIFORNIA, LOS ANGELES

Bachelor of Science in Cognitive Science (Specialization in Computation), Data Science Engineering Minor

Los Angeles

Expected June 2025

Cumulative GPA: 3.935/4.0, Cum Laude, Dean's Honor List, UCLA DataRes Research, UCLA Bruin Sports Analytics

RESEARCH & WORK EXPERIENCE

Tiktok Inc.

Machine Learning Engineer Intern

Seattle

May 2026 – Present

- Built an **end-to-end** teacher–student **distillation pipeline** for personalized e-commerce intent recognition using query signals and multimodal user behavior, covering streaming offline labeling, data infra, and student models optimized for **150ms** online serving
- Productionized an agentic labeling pipeline on daily-scheduled Spark, SQL, and LLM DAGs with behavioral-signal APIs and **concurrency** controls, generating 40K reasoning-annotated labels per day at **85% accuracy** on human alignment
- Architected a sparse-ID model combining SENet, DCN, LHUC personalized gating, and DIN sequence attention across 150+ sparse, dense, and sequential features; scaled training with class-weighting, parameter-servers, and streaming data ingestion
- Trained a 0.6B-parameter embedding-native **Qwen3 decoder** from scratch with per-token FFN feature encoders, Perceiver cross-attention, latent reasoning, and **RL post-training**; improved AUC from 0.52 to 0.83 and accuracy to 79%

Anima Lab, [Website](#)

Founder & Founding Engineer

Philadelphia

April 2026 – Present

- Built and launched a full-stack AI-companion product from 0→1 across Next.js web, native SwiftUI iOS, and React Native Android on a shared TypeScript/Convex backend; reached **1,000+ users**, recurring monthly revenue, and **40K+ followers**.
- Architected a real-time multimodal agent with photorealistic image generation, voice synthesis (MiniMax), and tool calling for web search on a **self-evolving** memory system: semantic retrieval (relevance–recency–importance reranking), consolidation cycles where the agent rewrites and prunes its own memory, and dynamic emotion/attachment state; raised **recall accuracy by 13%** and **MAU 20%**
- Led a **5+ person** product team and built cross-platform growth infrastructure and acquisition channels, increasing **signup rate by 15%**

Columbia DAP Lab (Data, Agents, Processes), Advisor: [Prof. Zhou Yu](#)

Machine Learning Researcher

New York

April 2025 – Dec 2025

- Fine-tuned Qwen2.5 and Qwen3 using RLHF framework, combining **SFT finetuning** and **GRPO** (VERL reward) to improve field-level structured generation in Elasticsearch-style job-resume retrieval; achieved **+17% nDCG** over ConFit-V2 baseline
- Built scalable SFT training and synthetic data generation pipeline using vLLM, LLaMA-Factory, and DeepSpeed ZeRO-3 on A100s
- Implemented a distillation-based data refinement pipeline (DeepSeek, GLM-4.7) with LLM judge re-ranking to denoise weak labels
- Trained GRPO using VERL with custom reward, curriculum strategy, and loose parsing design, yielding a **+23% recall improvement**.
- Applied one-shot learning and **self-rationalization** to infer recruiter-intent keywords, improving interpretability and **nDCG by 10%**

Job Junior

Co-Founder & Founding Engineer

Los Angeles

Sep 2023 – May 2026

- Built an end-to-end LLM-based outbound automation system for job and candidate matching; used **Playwright**, **LangChain**, **FAISS**, and **QLoRA** to extract job descriptions, match profiles, and generate cold emails with **15%+ response rate** across **1K+ targets**
- Built an asyncio-based ETL pipeline (MongoDB, FastAPI, AWS EC2/Docker) achieving **6x throughput**, enabling daily scraping of 800+ companies with automated delivery via lark and spreadsheet APIs; signed a **\$2K+/month** pilot with 5 top recruiting firms.

Generative Machine Learning Research Group, Advisor: [Prof. Jiatao Gu](#), [Paper link](#)

Machine Learning Researcher

Philadelphia

March 2024 – Present

- Engineered a multimodal LLM agent with **RAG memory** and self-reflection loop; supported language/image input to structured blueprint generation and autonomous execution via Node.js control bridge for Minecraft bots; **work published at AAAI W17**.
- Developed a verification pipeline combining multi-view visual feedback and world-state block scanning, enabling ground-truth structural diffing and iterative self-correction with actionable patch generation, outperforming one-shot generation accuracy by 25%
- Designed a trajectory-driven training pipeline leveraging one-shot agent rollouts and Blueprint DSL as supervision to fine-tune a Qwen3 Generator (SFT + RL), improving spatial consistency and structural correctness in generative construction tasks by **12%**

PROJECT EXPERIENCE

Synthetic Data Generation with LLM for Improved Depression Prediction, [Paper link](#)

Generative Data Science Grad Course

Los Angeles

April 2024 – July 2024

- Built a **CoT-style** LLaMA3 QLoRA model to generate privacy-preserving synopses from patient transcripts for PHQ-8 depression prediction; downstream **BERT regressor** achieved **RMSE 4.39 vs 5.01 baseline**, indicating improved diagnostic accuracy
- Evaluated fidelity via **PCA** on **CLS embeddings**, revealing strong structural alignment and representation of severe-depression cases
- Quantified privacy risk using embedding-distance metrics, achieving **min 4.96 / avg 7.11** vs baseline 4.04 / 5.91, demonstrating reduced information leakage; work accepted at the Annual Meeting of the Cognitive Science Society (CogSci 2024)